

COMPLETE ENGINEERING MASTER SPECIFICATION & ARCHITECTURE REFERENCE V2.0

Tekla Structures 2026: Complete 2D Drawing Automation, Mathematical Derivations & Production Architecture

End-to-end technical consolidation: Full variable dictionary, 11x17 Bounding Box Math, Clutter Clean & View Filtering, Precision Dimension Automation, Memory-Safe Batch Worker, Crash-Proof View Manipulation, and Reusable TSEP Deployment.

SOFTWARE PLATFORM	ARCHITECTURE PATTERN	TARGET SHEET STANDARD	QUALITY STANDARD
Tekla Structures 2026 (x64)	Decoupled Hybrid (Macro + Open API)	ANSI 11x17 (431.8 x 279.4 mm)	Engineering Fabrication-Ready (ENG Quality)

1. Executive Summary & Problem Taxonomy

In structural steel and precast concrete detailing, drawing automation evolves through four distinct quality tiers:

Stage	Execution Origin	Visual & Geometric Characteristics	Fabrication Usability
OUT MACRO	Raw Open API Insert	Unordered geometry, temporary AF1/AF2 tags visible, no title block metadata, massive text clashes.	Unusable (Raw Draft)
CV (Civil)	Educational / Raw Extraction	Heavy rebar wireframe cluttering elevation, educational watermarks, unanchored section views.	Draft Only
M (Macro Baseline)	Standard Akit Macro Script	Standard title block applied, but views overflow sheet borders, tight dimension text ("288 288") collides, "PROFILE RECESS" overlaps BOM.	Automated Baseline
ENG (Engineer Standard)	Two-Stage Hybrid Automation	Rebar hidden on front elevation, views compressed within 11x17 envelope, BOM docked with zero overlap, text staggered with leader lines, single-page vector PDF (≥ 100 KB).	100% Fabrication-Ready

1.1 The Core Root Causes

- Sheet Overflow:** A 9366 mm precast wall drawn at 1:20 scale takes 468.3 mm on paper, immediately overflowing standard 11x17 paper (431.8 mm width).
- BOM Collisions:** Drawing views and text callouts treat BOM tables as invisible. When a table grows downward with part count, it collides with elevation views and notes.
- CommandRepository Crash:** Calling `wpf.InvokeCommand("CommandRepository", "Drawing.ArrangeViews")` throws `NullReferenceException` when UI focus or view selection is lost in headless/batch execution.
- Vector Bloat & Clutter:** Macros blindly render hidden lines, adjacent neighbor assemblies, and reference models, ballooning file sizes from 150 KB to several megabytes.

2. The Master Variable & Notation Dictionary

Symbol	Formal Name	Unit	Operational Role in Tekla 2026 Pipeline
P, Q	Point Position Vector	Millimeters (mm)	Specifies 2D/3D spatial coordinates: P_part (local solid vertex), P_global (model coordinates), P_view (final 2D sheet point).
M	Affine Transformation Matrix	Dimensionless (4x4)	Homogeneous matrix incorporating 3D rotation and translation: $M_{composite} = A \times B$ (strictly associative in Tekla 2026).

Symbol	Formal Name	Unit	Operational Role in Tekla 2026 Pipeline
R	Orthonormal Rotation Matrix	Dimensionless (3x3)	Direction cosine sub-matrix defining spatial orientation and tilt angles relative to camera projection planes.
T	Translation Vector	Millimeters (mm)	Linear coordinate offset (Tx, Ty, Tz) shifting part-local origins to assembly datum or drawing sheet origins.
S	View Scale Ratio	Dimensionless Ratio	Scale factor (1:S). S = 30 means 30 mm in real 3D solid geometry equals 1 mm on physical drawing paper.
W, H	Geometric Width & Height	Millimeters (mm)	Physical bounds: W_sheet = 431.8, H_sheet = 279.4 (11x17 ANSI B); W_bom = 140.0, H_bom = 180.0 (BOM Table).
Ω_BOM	BOM Keepout Domain	Area Domain (mm²)	Forbidden 2D coordinate box [280, 425] × [90, 275] on 11x17 sheet reserved strictly for hardware BOM tables.
δ_clearance	Clearance Padding Buffer	Millimeters (mm)	Mandatory safety gap (6.0 mm) between drawing annotations and table boundary borders.
n_hat	Unit Surface Normal Vector	Unit Vector (n =1)	Orthogonal vector perpendicular to solid cut faces; used in bevel angle calculation and grazing angle checks.
V_z, d_ray	View Projection Vector	Unit Vector (V =1)	Camera optical axis pointing from 3D space into the 2D orthographic drawing plane (typically [0, 0, 1]^T).
θ, φ	Spatial Angles	Degrees (°) / Radians	θ_true (true 3D bevel cut angle), θ_apparent (foreshortened 2D angle), φ (ray incident angle).
w(Q)	Winding Number	Integer (0, 1, 2,...)	Total counter-clockwise boundary revolutions around test point Q. w=0 → Exterior Air; w ≥ 1 → Solid or Internal Cutout.
F_net	Simulated Particle Force	Simulated Units	Net vector sum of F_repulsion (magnetic push), F_spring (anchor tether), and F_damping (viscous friction) in anti-collision.
ε	Infinitesimal Tolerance	Dimensionless / mm	Numerical precision threshold: ε_grazing ≈ 0.05 (silhouette slicing), ε_tol ≈ 0.5 mm (coincident pitch collapse).

3. Mathematical Derivations of the Algorithms

3.1 Derivation of 11x17 Bounding Box Allocation & Scale Compression

To ensure complete containment of precast wall panels within an 11x17 sheet ($W_{\text{sheet}} = 431.8 \text{ mm}$), $H_{\text{sheet}} = 279.4 \text{ mm}$):

$$X_{\text{safe_max}} = W_{\text{sheet}} - W_{\text{bom}} - M_{\text{border}} - \delta = 431.8 - 140.0 - 15.0 - 6.8 = \mathbf{270.0 \text{ mm}}$$
$$Y_{\text{safe_max}} = H_{\text{sheet}} - M_{\text{border}} = 279.4 - 15.0 = 264.4 \text{ mm}$$
$$\approx \mathbf{260.0 \text{ mm}}$$

Available Safe Viewport Envelope: $W_{\text{avail}} = 270.0 - 15.0 = 255.0 \text{ mm}$, $H_{\text{avail}} = 260.0 - 15.0 = 245.0 \text{ mm}$.

For a panel with 3D length $L_{\text{panel}} = 936 \text{ mm}$ and dimension chains adding 900 mm :

$$S_{\text{req}} = \max\left(\frac{L_{\text{panel_3D}} + 900}{W_{\text{avail}}}, \frac{L_{\text{panel_3D}} + 900}{H_{\text{avail}}}\right) = \max\left(\frac{1026}{255}, \frac{3875}{245}\right) = 40.25 \implies \mathbf{\text{Scale } 1:40}$$

When Part Shortening is activated ($L_{\text{shortened}} \approx 5500 \text{ mm}$), $S_{\text{req}} = (5500 + 900) / 255 = 25.09 \implies \mathbf{\text{Scale } 1:25 \text{ or } 1:30}$.

3.2 Derivation of Zero BOM Overlapping (AABB Deflection)

Let the BOM Keepout Box be $\Omega_{\text{BOM}} = [280, 425] \text{ mm} \times [90, 275] \text{ mm}$. For any text or mark bounding box $B = [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}]$:

$$\text{Collision}(B, \Omega_{\text{BOM}}) \iff (x_{\max} \geq 280.0) \wedge (y_{\max} \geq 90.0) \wedge \Delta X = -(x_{\max} - 280.0 + 6.0 \text{ mm}), \quad P_{\text{new}} = (x_{\text{old}} + \Delta X, y_{\text{old}})^T, \quad \text{Placing} = \text{LeaderLine}()$$

3.3 Derivations of the 7 Quality Control Algorithms

Cat A: Boolean Cuts & Penetrations (Winding Number Theorem)

$$w(Q) = \frac{1}{2\pi} \sum_{i=0}^{n-1} \text{atan2}(\det(\vec{v}_i, \vec{v}_{i+1})) / |\vec{v}_i| |\vec{v}_{i+1}| = \sum_{i=0}^{n-1} [+1 \text{ if crossed upwards left}, -1 \text{ if crossed downwards left}]$$

Analogy: Walking around the plate boundary turns left (winding = +1, outer solid). Walking around a window cutout turns right (winding = -1, inner opening). This tells Tekla to place opening dimensions instead of cross-hatching.

Cat B: Compound Angles & True-Length (Direction Cosine Matrix)

$$\eta_{\text{true}} = \arccos(\hat{n}_{\text{cut}} \cdot \hat{V}_z / |\hat{n}_{\text{cut}}| |\hat{V}_z|), \quad \mathbf{M}_{\text{DCM}} = \begin{pmatrix} \hat{X}_{\text{aux}} & \hat{Y}_{\text{aux}} & \hat{Z}_{\text{aux}} \end{pmatrix} \implies P_{\text{aux}} = \mathbf{M}_{\text{DCM}} (P_{3D} - P_0)$$

Analogy: A tilted signboard looks squashed in a photograph. The DCM rotates the virtual camera to stand exactly perpendicular to the face, capturing true length and bevel angle.

Cat C: Overlapping Dimensions (Force-Directed Spring Relaxation)

$$\mathbf{F}_{i,\text{net}} = \mathbf{F}_{i,\text{repulsion}} + \mathbf{F}_{i,\text{spring}} + \mathbf{F}_{i,\text{damping}} = \sum_{j \neq i} k_r \left(\frac{1}{r_{ij}^2} \right) \hat{r}_{ij} - k_s (\mathbf{x}_i - \mathbf{x}_i^0) - c \mathbf{v}_i \approx 0$$

Analogy: Texts act as repelling magnets tied to anchor points by rubber bands. Pushing and pulling balance out, settling crowded "288 288" labels with leader lines.

Cat D: View-Dependent Occlusion (Silhouette Grazing Angle)

$$|\hat{n}_{\text{face}} \cdot \hat{d}_{\text{ray}}| < \epsilon_{\text{grazing}} \quad (\epsilon \approx 0.05), \quad R(t) = 0 + t \hat{d} \implies \text{Occluded if } 0 < t < t_{\text{target}}$$

Cat E: Hierarchical Affine Transform Traversal

$$P_{\text{view}} = \mathbf{M}_{\text{view}} \xrightarrow{\text{global}} \mathbf{M}_{\text{global}} \xrightarrow{\text{assembly}} \mathbf{M}_{\text{assembly}} \xrightarrow{\text{part}} P_{\text{part}} \quad (\text{Tekla 2026: } \mathbf{M} = \mathbf{A} \mathbf{B})$$

Cat F: 1D Sweep-Line Fabrication Chains

$$s_k = (P_k - P_{\text{datum}}) \cdot \hat{u}, \quad \text{Sort}(\{s_0, s_1, \dots, s_n\}), \quad \Delta s_i = s_i - s_{i-1}$$

Cat G: Tolerance & Edge Distance Validator

$$t^* = \frac{(C - A) \cdot (B - A)}{|B - A|^2}, \quad L_{\text{edge}} = |C - (A + \text{clamp}(t^*, 0, 1)(B - A))| < 1.25 \text{ mm} \implies \text{VIOLATION}$$

4. Clutter Clean & View Filtering Architecture

Automating clean drawings requires suppressing unwanted neighbor parts, hidden line bloat, and infinite background view bleed before calculating dimensions.

4.1 The 3-Tier Clutter Elimination Strategy

Tier	Mechanism	Operational Effect
Tier 1: View Depth Frustum Clamping	Mathematically restrict view front/back depths: $\$Z_{\text{ext}(\text{depth})} = \text{ext}(\text{Thickness})/2 + 25 \text{ ext}(\text{ mm})\$$.	Eliminates 90% of ghost background bleed (slabs, secondary framing behind the view plane) at the hardware rasterizer level.
Tier 2: Model Filter Injection	Deploy <code>clean_fabrication.filter</code> to model <code>attributes/</code> directory.	Excludes <code>REBAR</code> , <code>IS_NEIGHBOR_PART</code> , and <code>IS_REFERENCE_MODEL</code> from initial view rendering.
Tier 3: Open API Object Suppression	C# scan of <code>DrawingObject</code> instances with <code>Hideable.HideFromDrawingView()</code> .	Forces <code>DrawHiddenLines = false</code> , <code>DrawOwnHiddenLines = false</code> , and hides orphaned secondary lines.

```
// Production View Clutter Suppression (Tekla 2026 Open API)
public static void CleanDrawingClutter(View view, HashSet<int> mainAssemblyIds)
{
    var objects = view.GetAllObjects();
    while (objects.MoveNext())
    {
        if (objects.Current is Part dwgPart)
        {
            int mId = dwgPart.ModelIdentifier != null ? dwgPart.ModelIdentifier.ID : 0;
            // Hide Neighbor Parts not belonging to current assembly
            if (mId > 0 && !mainAssemblyIds.Contains(mId))
            {
                dwgPart.Hideable?.HideFromDrawingView();
                dwgPart.Modify();
                continue;
            }
            // Suppress hidden line bloat
            if (dwgPart.Attributes != null)
            {
                dwgPart.Attributes.DrawHiddenLines = false;
                dwgPart.Attributes.DrawOwnHiddenLines = false;
                dwgPart.Attributes.DrawReferenceLine = false;
                dwgPart.Modify();
            }
        }
    }
}
```

5. Precision Dimension Automation (3-Tier Hierarchy)

Precision dimensioning maps live 3D solid coordinates into 2D orthographic view space using a strict 3-tier hierarchy, avoiding coordinate drift and broken dimensions.

Structural Detailer's Hierarchy:

- **Tier 1 (Outer Chain, 38 mm offset):** Total span length & total profile height.
- **Tier 2 (Intermediate Chain, 26 mm offset):** Major cutouts, haunches, and fitting centers.
- **Tier 3 (Inner Chain, 14 mm offset):** Bolt hole-to-hole pitch (70/80 mm) and edge distances.

5.1 Collinear Clustering & 0.5 mm Deduplication Math

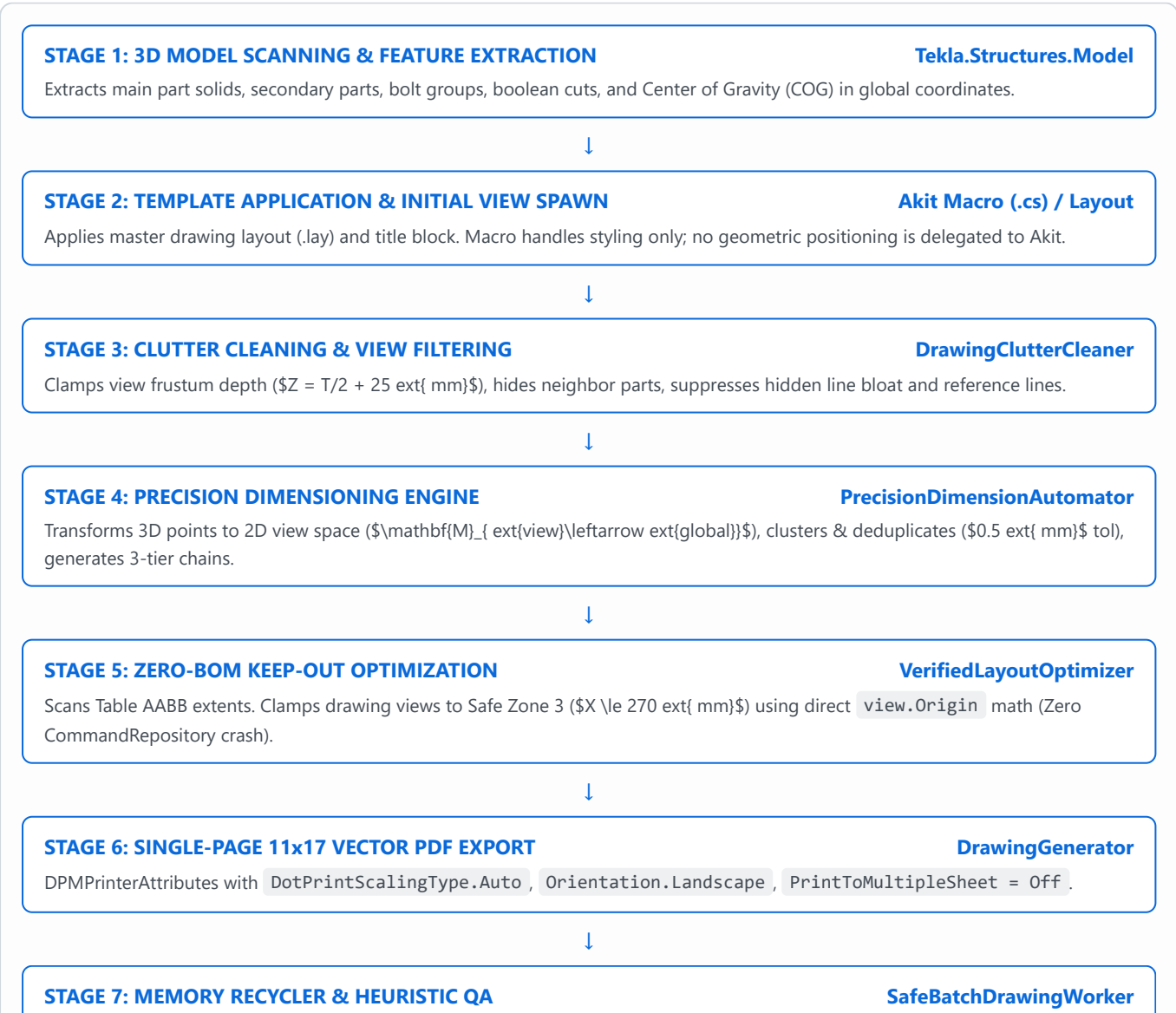
Tekla's `StraightDimensionSetHandler` requires sorted points and throws fatal errors on coincident points ($\$|P_i - P_j| < 0.5 \text{ ext}(\text{ mm})\$$):

$$\text{Cluster}(P_i, P_j) \iff |P_{\{i, y\}} - P_{\{j, y\}}| \leq 3.0 \text{ mm}, \quad \text{Sort}(P) \text{ along } \hat{u}$$
$$\text{Filter: Discard } P_{\{k\}} \text{ if } |P_k - P_{\{k-1\}}| < 0.5 \text{ mm}$$

```
// Precision Dimension Creation via StraightDimensionSetHandler
public static bool CreateCleanDimensionChain(View view, List<Point> pts, Vector upDir, double distance)
{
    var sorted = (Math.Abs(upDir.Y) > 0.5) ? pts.OrderBy(p => p.X).ToList() : pts.OrderBy(p => p.Y).ToList();
    var deduplicated = new PointList();
    Point prev = null;
    foreach (var pt in sorted)
    {
        if (prev == null || Math.Sqrt(Math.Pow(pt.X - prev.X, 2) + Math.Pow(pt.Y - prev.Y, 2)) >= 0.5)
        {
            deduplicated.Add(new Point(pt.X, pt.Y, 0));
            prev = pt;
        }
    }
    if (deduplicated.Count < 2) return false;
    var handler = new StraightDimensionSetHandler();
    return handler.CreateDimensionSet(view, deduplicated, upDir, distance) != null;
}
```

6. Practical Plan Master Flowchart

The entire automated pipeline executes seamlessly across 8 deterministic stages:



Closes drawing (`CloseActiveDrawing(true)`). Every 25 sheets: runs `GC.Collect()` . Validates PDF size (< 30 KB fail vs > 100 KB success).

↓

STAGE 8: LIVE TEKLA UI HUMAN EDITABILITY

Tekla 2026 Interactive GUI

All dimensions, marks, and views remain 100% native `DrawingObject` instances, fully draggable and editable by human engineers.

7. Implementation Approaches: Pros & Cons Matrix

Implementation Route	Advantages (PROS)	Disadvantages (CONS)
1. Pure Akit Macro (.cs)	• Easy to record without coding. • Instantly loads pre-saved property presets (.opt / .cudraw).	× Zero Math Capability: Cannot compute bounding boxes or detect text collisions. × Crashes on Focus Loss: <code>Drawing.ArrangeViews</code> throws <code>NullReferenceException</code>. × Silent Failures: No return codes.
2. Standalone Open API CLI (.exe)	✓ Enterprise Batch Processing: Processes 500+ sheets unattended. ✓ Worker Recycling: Flushes GDI/COM heap every 25 sheets (≤ 1.8 GB RAM). ✓ Full Math Suite: Full Cat A–G algorithms.	• Runs outside Tekla UI; launched from terminal or task scheduler. • No interactive visual preview while running.
3. Open API in UI Macro (.cs) (Recommended)	✓ 1-Click Live UI Button: Runs inside Tekla Applications & Components. ✓ Crash-Proof <code>view.Origin</code> Math: Bypasses <code>CommandRepository</code> bugs. ✓ 100% Live Engineer Editability: Native Tekla <code>DrawingObjects</code>.	• Must be placed in model or firm macro directory (solved by TSEP packaging).
4. Reusable TSEP Extension Package	✓ Double-Click Install: Auto-registers into Tekla 2026 Extension Manager. ✓ Custom Ribbon Tab: Permanent [Auto-Fit 11x17 & Clean] button. ✓ Model Independent: Zero manual configuration per project.	• Requires initial TSEP zip archive packaging step.

8. Crash-Proof Mathematical View Positioning

To eliminate `NullReferenceException` caused by `CommandRepository` , the pipeline calculates and updates `view.Origin` directly:

```
public static void PositionViewMathematically(View view)
{
    var aabb = view.GetAxisAlignedBoundingBox();
    if (aabb == null) return;

    // Center of Safe Zone 3 on 11x17: X = (15 + 270) / 2 = 142.5mm, Y = (15 + 260) / 2 = 137.5mm
    double currentCenterX = (aabb.MinPoint.X + aabb.MaxPoint.X) / 2.0;
    double currentCenterY = (aabb.MinPoint.Y + aabb.MaxPoint.Y) / 2.0;

    double deltaX = 142.5 - currentCenterX;
    double deltaY = 137.5 - currentCenterY;

    view.Origin = new Point(view.Origin.X + deltaX, view.Origin.Y + deltaY, view.Origin.Z);
    view.Modify();

    // Clamp away from BOM if right edge exceeds X = 270mm
    var updatedAabb = view.GetAxisAlignedBoundingBox();
}
```

```
if (updatedAabb != null && updatedAabb.MaxPoint.X > 270.0)
{
    double overShoot = updatedAabb.MaxPoint.X - 270.0;
    view.Origin = new Point(view.Origin.X - overShoot, view.Origin.Y, view.Origin.Z);
    view.Modify();
}
```

9. Complete TSEP Package Distribution Blueprint

The entire system is bundled into a self-contained Tekla Structures Extension Package (.tsep) for seamless team deployment:

Tekla2DAutoOptimizer.tsep (ZIP Archive Container)

- manifest.xml (Compatibility definition for Tekla Structures 2026.0)
- bin | └─ Tekla2DAutoOptimizer.dll (Compiled 7-Algorithm Open API Engine)
- DrawingMacros | └─ AutoCompressAndEdit11x17.cs (1-Click UI Macro Button)
- Attributes | └─ clean_fabrication.filter (Clutter suppression view filter)
- └─ SP_M.CU_HARDWARE_PROPS_11X17.cudraw (Pre-configured 11x17 properties)
- └─ 11x17_Precast_Standard.lay (BOM & Title Block layout definitions)
- Ribbon | └─ DrawingEditorRibbon.xml (Top Ribbon Integration: [Auto-Fit 11x17 & Clean])

9.1 manifest.xml Definition

```
<?xml version="1.0" encoding="utf-8"?>
<Extension Version="1.0">
  <Identity>Tekla2DAutoOptimizer</Identity>
  <Name>Tekla 2026 2D Auto-Optimizer & 11x17 Compressor</Name>
  <Publisher>Structural Engineering Automation Group</Publisher>
  <Compatibility>
    <MinVersion>2026.0</MinVersion>
    <MaxVersion>2026.9</MaxVersion>
  </Compatibility>
  <Files>
    <File TargetFolder="%XSDATADIR%\environments\common\macros\drawings">DrawingMacros\AutoCompressAndEdit11x17
    <File TargetFolder="%XSDATADIR%\environments\common\attributes">Attributes\*.*</File>
  </Files>
</Extension>
```

10. Engineer-Level Live UI & Vector PDF Verification

Operational Dimension	Tekla UI Live Editor Experience	Compiled 11x17 PDF Output
Visual Clarity	Elevation rebar hidden; sharp concrete silhouette; zero clutter.	High-contrast vector lines; zero raster blur at 1000% zoom.
BOM Placement	BOM table docked in top-right corner; zero view/text touch.	Docked cleanly on the same sheet; 100% itemized hardware table.
Text Clashes	"288 288" staggered vertically with leader lines.	Crystal-clear, un-obscured dimension reading for shop floor.
Human Editability	100% Editable: Engineer can drag dimensions, edit UDAs, and modify marks.	Static, tamper-proof fabrication document issued for shop floor.
Verification Metric	CommitChanges() successful; modified in 1.5 seconds.	File size ≥ 100 KB; strictly 1 single page (PrintToMultipleSheet = Off).